

Implementation of Hierarchical Oblivious Stack in EMP

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Abstract

Implementing stack functionality in secure computation is nuanced. A naïve MPC implementation of stack requires access time linear in the maximum size of the stack N (i.e. entire encrypted stack container must be scanned). A much better approach, with access cost $O(\log N)$, was proposed by Zahur et al. (IEEE S&P 2013).

In this paper, we introduce and motivate the problem of implementing oblivious stack in MPC, describe the protocol of Zahur et al., and report on our experience of implementing it using EMP toolkit (Wang et al.).

1 Outline

This work details experiences in learning and implementing oblivious stack [4] with the EMP toolkit [3]. The source code of our project is available [2]. Our presentation is structured as follows:

We begin in Section 2 by introducing oblivious stack, its motivation, and highlighting its differences from the standard (non-private) stack implementation. We introduce the EMP toolkit, which we use as a framework for our implementation.

Then, Section 3, Oblivious Stack Design, will walk through all the functions of the oblivious stack and explain how they work and why they are used.

Third, Section 4, Implementation, will walk through the data structures used to implement the stack and how each of the functions are implemented practically.

Finally, we discuss the performance of our implementation and noteworthy implementation details in Section 5.

2 Introduction

A stack is a linear data structure that follows the Last-In, First-Out (LIFO) principle, meaning the last element added to the stack is the first one to be retrieved (and removed from stack). These operations are implemented by `push` and `pop` functions.

However, implementing it in secure computation, or MPC (Multi-Party Computation), is somewhat challenging. MPC must hide all intermediate values of the execution. Thus all internal values, as well as the program flow that depends on internal values, must be hidden from all parties executing the program.

As it turns out, hiding the program flow is the most challenging aspect of MPC. A popular way to run MPC is by executing programs represented as circuits, e.g. Boolean circuits. This is because the program flow is preset so the encrypted internal values do not need to be known to manage the flow of the program. However, this comes with a major downside: conditionals are extremely expensive in circuits. To implement a conditional in a circuit (either a plaintext circuit or an MPC execution), both branches of the conditional must be executed. The result is then selected according to the value of the selection wire.

This approach leads to exponential growth of execution time if nested conditionals occur. A good trade off in MPC is to have straightline programs (efficiently implementable as circuits), that can operate with conditional/oblivious data structures. Note, this approach may not *always* convert a function to a small circuit, but it may improve performance in many functions. In our case (secure computation), it is useful to have a stack whose operations (`push` and `pop`) take an additional Boolean parameter that controls whether or not the operation is actually performed.

EMP [3] is a toolkit that allows users to write MPC programs in C++. It compiles C++ programs into circuits that can be run efficiently. EMP handles conditionals as described above: by generating and executing both branches. In this paper, we are concerned with implementing the stack efficiently in MPC using functions and data structures contained in EMP. It is convenient for us to introduce the notation of *oblivious* values, such as oblivious integer or oblivious Boolean. These are the values that are not known to either party during evaluation, but are jointly known because they are secret-shared in some way. EMP extensively uses oblivious values; it includes classes `Integer`, `Bit`, etc. that implement oblivious integer, bit, etc. In programming MPC, we need to operate with both plaintext and oblivious values.

3 Oblivious Stack Design

`condPop` (conditional `pop`) and `condPush` (conditional `push`) are the main func-

tions of the stack: they are what the user calls in order to **push** and **pop** objects. **condPush** takes in the value being **pushed** (an oblivious integer) and an oblivious bit, and returns nothing. **condPop** takes in an oblivious bit and returns the **popped** value (an oblivious integer). Internally, if **condPush** is called with a false bit, it mechanically inserts the **pushed** value into the stack (to hide the fact that no insertion takes place), but does not change the (oblivious) pointers that keep the order of the stack. This is done by, e.g., adding an oblivious value 0 to the pointers). This effects no change to the actual content of the stack, but follows the control flow that is identical to the one where the element is logically inserted. Similarly, if **condPop** is called with a false bit, it returns the value at the top of the stack but does not change the pointers, keeping the top value in the stack.

```
condPush(True, 5); //will push 5 to the stack
condPush(False, 3); //will not change the logical content of the stack
condPop(True); //will correctly pop
condPop(False); //will not change the logical content of the stack
```

Figure 1: Illustration of the **condPush** and **condPop** function calls

A naïve way to implement oblivious stack is to simply store it in an array and keep an oblivious pointer to the top of the stack. However, this is inefficient: because we are implementing the stack obliviously, we would need to scan the entire stack every time we accessed it to avoid revealing an access pattern to potential attackers. Additionally, because stack operations are oblivious, no player actually knows the state of the stack. That is, the stack’s size, top location, etc., are all encrypted. Therefore, there is no way of executing **condPush** or **condPop** with this approach without accessing every single element.

An elegant solution to this is to store the stack in a *hierarchical* data structure, allowing us to scan an array of only a small constant size when we access the stack. This method was proposed in [4]. We closely follow their algorithm.

That is, we represent the stack as a collection of arrays where each successive array is multiplicatively (twice, in our algorithm) larger than the previous one (thus array size grows exponentially in the collection). The stack is organized so that the elements of the top of the stack are always contained in the first (top) array. This allows all of the **push** and **pop** operations to complete by scanning only the first array. (As we explain later, it is sufficient to store only 5 elements in this array). This means that every time we **push** or **pop**, we only need to scan an array of size 5, instead of scanning the entire stack storage. However, because we only **push** and **pop** into/from the top array, we need a method of distributing, storing and accessing the values in the lower levels. The functions

`shiftRight` and `shiftLeft` take care of this¹: `shiftRight` and `shiftLeft` are efficient functions that keep each level in a correct range of occupancy so that we can `push` and `pop` from the first level only. They ensure that if the top array is close to overflowing, the data gets moved to lower larger arrays (by calling `shiftRight`), and conversely `shiftLeft` moves the data up when needed so that it is always available in the top array for a `pop`. The challenge is that these functions must operate obliviously, without plaintext knowledge of the stack access operations. This is because the information about the stack is all encrypted and cannot (cheaply) be used as a conditional. A key idea, introduced in [4] is that this can be done obliviously and efficiently according to a fixed schedule of movement operations, described next.

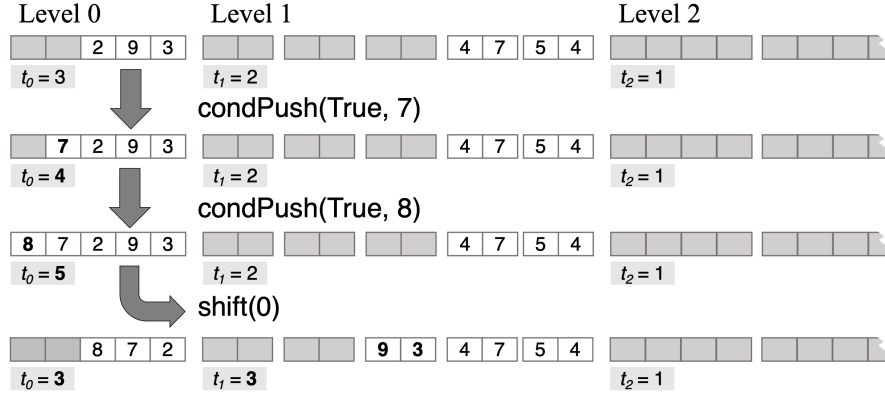


Figure 2: Visual representation of the `shiftRight` function on level 0. `shiftRight(0)` is called after every 2 `condPush` operations, as long as level 0 contains 3 or more values.

`shiftRight` is a function that shifts blocks of elements down from the higher levels to the lower levels according to a fixed schedule of movement operations. It is demonstrated in Figure 2. Recall, obviously accessing an array element involves scanning the entire array (and executing in MPC a corresponding circuit of size proportional to array size). The key to the efficiency of the hierarchical stack lies in the fact that we shift *in blocks*. The improvement comes from needing to traverse the array once for each *block* (of elements) as opposed to once for each element. Indeed, element j of a block is always read from and written to the j -th position after the first element of the block, eliminating the need to scan the array for any block element except for the first one.

¹In this paper we interchangeably use the notation from [1] (a concise and intuitive overview of main MPC constructions; [1] orders hierarchical arrays left to right, with the left being smallest) and [4], which orders top down, with the top array being smallest. Thus, e.g., our `shiftRight` function shifts data to the lower, larger, array.

For level i , `shiftRight` shifts block of size 2^{i+1} elements down to level $i + 1$. For example, `shiftRight` moves 2 *consecutive* elements (as one block!) down from level 0 to level 1, and 4 such elements down from level 1 to level 2. To highlight the savings from block-based movement, notice that if we shift right a block from level 10 to level 11, we traverse level 10 only once, instead of 2,048 times of if we moved elements individually.

`shiftRight` is called on a schedule: for a level i it is called after every 2^{i+1} pushes. It obviously moves the data to the lower array if such a move is needed (i.e. the higher array is getting full). `shiftLeft` schedule, operation and tasks are symmetric to `shiftRight`.

Crucially, this algorithm ensures that the stack can accept any sequence of pushes and pops. This is because shifting will move elements *to* a level (from below) when it is too empty after popping or shifting, and shift elements *away* (down) when the level is too full after pushing or shifting. Indeed, the maximal data movement rate provided by shifting is greater than that of push and pop operations: in the worst case, a sequence of K true pushes (i.e. pushes called with true bits) pushes K elements. At the same time `shiftRight` will be called at least $\lfloor K/2^i \rfloor$ times for each level i , moving 2^i elements with each shift. $K/2^i \cdot 2^i$ is equal to K , meaning that for every K pushes, each level will shift around K elements down. This keeps the number of elements in each level stable. Similarly, this is also the case for K pops in a row.

The performance of our implementation of this algorithm is analyzed in Section 5.

4 Implementation

In this section, we discuss lower-level implementation details and decisions that we made.

We used a Standard Template Library (STL) `Vector` of `IntArray`s of increasing size as the hierarchy of storage arrays. We will have $\lceil \log(N) \rceil$ of arrays to support a stack of up to N elements. An `IntArray` is a vector of `Integers` (oblivious integers). The `Integer` class is included in EMP. It includes mathematical operations, bit-wise operations, and additional efficient functions that can help with writing programs. `IntArray` is an auxiliary class written by us that implements an oblivious array of `Integers`, with corresponding accessor functions. `IntArray`s can be indexed and written to using both plaintext and oblivious indices. Accessing and writing using plaintext indices takes $O(1)$ time, and using oblivious indices takes $O(N)$ time, where N is the length of the array. Each `IntArray` represents a level of the stack.

An essential part of the algorithm are the pointers that allow us to know where the head and tail of each level are. We create three `IntArray`s of `Integers` (i.e.

EMP oblivious integers): Heads, Tails, and Lengths. These arrays are the same length as the number of levels in the stack. For each level i , Heads[i] contains the index of the head of the level, or the element in the level that was **pushed** first chronologically. Tails[i] stores the index of the tail of the level, or the first empty slot in the level. That is, Tails[0] points to the empty slot on the top of the stack. Lengths[i] contains the number of elements currently stored in level i . Heads, Tails, and Lengths are sufficient to allow us to perform calculations to get the elements we need for **pushes** and **pops**. Sometimes when shifts are called, the elements need to "wrap around" the array in order to fit; to account for this, Heads and Tails are (obliviously) modded by the length of the array.

Now, we will walk through each function. First, **condPush**. The **condPush** operation always pushes into the top level. To **condPush**, the (oblivious) value is inserted at the index contained at Tails[0] (which points to the next available empty slot on the top of the stack). The oblivious bit B passed into **condPush** is then added to the values at Tails[0] (mod 5) and Lengths[0]. This works because, if B is 1, a value will be inserted at the end of the first level. This means that Tails[0] needs to be updated by one to represent the next empty space in the stack, and Lengths[0] needs to be updated to represent the new length of the array. Otherwise, if B is 0, 0 will be added to Tails[0] and Lengths[0], and so the stack will not be changed. Indeed, the location of the value that is written will still be seen as empty: Tails[0] will still point to that location as the first unallocated spot.

Like the **condPush** operation, the **condPop** operation always **pops** from the top level. To **condPop**, we subtract B from Tails[0] and Lengths[0]. If $B = 1$, i.e. we are **popping**, both values will be reduced by one (mod 5 for Tails[0]). If B is 0, i.e. we are not **popping**, Tails[0] and Lengths[0] will remain the same. The **pop** operation always returns the Integer on the top of the stack, even if $B = 0$.

There are two global variables in the stack data structure: **push_counter** and **pop_counter**. These variables are plaintext integers that are increased by one each time **condPush** and **condPop** are called respectively. They are used to run the scheduled oblivious shifts of the data items. For each level i in the stack, **shiftRight** is called if **push_counter** mod 2^{i+1} is equal to 0. This ensures that **shiftRight** is called once for every 2^{i+1} **pushes**. Equivalently, **shiftLeft** is called if **pop_counter** mod 2^{i+1} is equal to 0.

Efficient Oblivious Shifting. We now describe our implementation of the procedure to efficiently obliviously shift (move between our hierarchical arrays) a *block* of elements, with only one scan of each of the source and destination arrays. Remember that each level has 5 equally sized blocks. For each block j in level i , we (obliviously, controlled by an oblivious bit set to 1 only for the blocks needed to be moved) copy the j th and $j+1$ st blocks to an auxillary array **aux** that holds exactly 2 blocks. This is done with a single pass over the source

array at level i at the cost of $O(2^i)$ operations². `aux` is then copied obliviously in the same fashion³ into level $i + 1$, completing the shift.⁴

At the implementation level, oblivious element moves between arrays are achieved using calls to a `IntArray` member function `obliviousWrite`. `obliviousWrite` takes in a plaintext index in the destination array⁵, an oblivious bit, and an `Integer` (oblivious integer) value, and writes the `Integer` into the the array it is called on, in the position of the index that is passed in. We emphasize, `obliviousWrite` writes directly to the destination index, and does not scan the destination array.

`shiftLeft` performs the same process symmetrically: it shifts from the top of array $i + 1$ to the bottom of array i .

5 Performance

The naïve stack design, where the stack is simply stored in an array-like data structure with an oblivious pointer to the top of the stack, is prohibitively slow. For all circuit-based MPC protocols, the primary cost scales linearly with the number of gates. In a stack implementation by a circuit, the cost depends on the maximum possible number of elements in the stack. Because the stack is oblivious and we don’t want to reveal the access pattern to potential adversaries, the naïve implementation scans the entire array of size N for every `push` or `pop`. Therefore, its total cost is $O(k \cdot N)$, where k is the number of `push` or `pop` operations. This could work for algorithms that handle small amounts of data, but would be too slow otherwise.

Our implementation of the stack, however, runs a lot faster, with an efficiency of $O(k \cdot \log N)$. This can be seen with the following calculation: Each level i is accessed at most $k/2^i$ times, because we perform a right shift into level i after every 2^i `pushes` and a left shift after 2^i `pops`. However, because the size of the level increases exponentially with i , so does the cost of shifting from that level. The size of each block at level i is 2^i , meaning we need $O(2^i)$ logic gates to shift it. Shifting large blocks at once (instead of shifting their elements individually) greatly decreases the cost – see “Efficient Oblivious Shifting” in Section 4 for discussion. Therefore, we need $O(k/2^i \cdot 2^i) = O(k)$ gates to operate shifting at each level. With $O(\log N)$ levels and $O(k)$ gates needed for each level, the total cost is $O(k \cdot \log N)$. This is significantly faster than the naïve version, especially as the size and number of elements increase.

²More precisely, each element of the source array is accessed twice – once as part of block j and once as part of block $j + 1$.

³The data move operation here is symmetric: the difference is that we iterate over the pairs of blocks of the destination array at the level $i + 1$.

⁴We use an auxillary array to avoid a nested `for` loop – in a naïve implementation, we would need to iterate through every block of level $i + 1$ for every block in level i to ensure that we are writing in the correct place, since both source and destination pointers are oblivious).

⁵Notice, we know the destination index because we write out the two blocks sequentially.

Notes on implementation, performance and measurement.

When I first wrote the code, it ran suspiciously quickly, at a sub-linear speed with very few gates. When oblivious code runs faster than expected, it often signifies improper encoding or encryption. In our case, the `Integers` we were using to `push` were `PUBLIC`, meaning that they were not encrypted for either Alice or Bob. This allowed the compiler to optimize out a lot of operations. When I switched the `Integers` to being encrypted on Alice’s side, the code ran at a more expected speed.

However, the runtime grew almost linearly for the input sizes I experimented with. The reason for this was that the linearly growing component of the code was implemented inefficiently, and greatly overpowered the component that scaled with $k \log N$.

To try to decrease the linear part of our code, I made some tweaks to the code for efficiency: for example, I noticed that the oblivious `mod` function was expensive because it performs division. To solve this, I created our own oblivious `mod` function using subtraction instead of division. This relies on the fact that in our protocol, the numbers I am performing the `mod` operation on are never more than twice the `mod` value. This decreased the number of gates in the resulting circuit by a factor of three.

Additionally, I realized that oblivious multiplication was a large cost and I was performing the same multiplication multiple times inside `for` loops. To fix this, I created a variable with the multiplication result and used it instead of performing the operation every time. This additionally decreased the number of gates in the resulting circuit by a factor of three.

After fixing these two things, the linear part of our code decreased significantly, making it fit the $k \log N$ model much better. However, the code was still running very slowly. Eventually I discovered the problem. By mistake, I was running the code by instantiating `EMP PlainProt` as `ProtocolExecution`. This runs the protocol in plaintext and writes the circuit generated from the code to a file. (This is by mistake; I used `PlainProt` to generate the circuits to measure their size, and forgot to revert to proper GC execution for evaluation). The circuit files are very large, and writing to files is slow, so my code ran slowly. When I switched to running the `SemiHonestParty` mode, my code ran almost 100 times as fast! This is because the GC data was quickly in-memory (I believe, bypassing the network stacks, because the two processes ran on the same machine) delivered to the EMP Evaluator process, which quickly processed and discarded it. The large circuit was never written to the (slow) disk.

Still, there was one final quirk. When I tested my code output after running it using `semi_honest`, I realized it again did not match well the expected $k \log n$ asymptotics. To fix this, I removed (a significant amount of) debugging output to the screen I performed during the execution. After removing this output, my

code ran much faster yet, and matched the $k \log N$ model much better.

5.1 Performance Report

To test our implementation, I performed p pushes and p pops on a stack instantiated to hold p elements. All tests were run on a MacBook Air laptop with an Apple M3 4.05 GHz processor and 16GB of RAM. Here were our results:

Input Size (p)	Number of Gates	Number of AND Gates	Time (seconds)
160	11,864,911	3,296,248	0.37
320	25,158,183	7,096,392	0.69
640	52,793,535	15,151,744	1.45
1280	110,130,647	31,497,365	3.05
2560	228,906,479	65,696,159	6.41
5120	474,630,151	136,218,853	13.44
10240	982,390,303	281,946,017	27.23
20480	2,030,505,018	582,754,940	59.00

Table 1: Number of gates in the circuit and execution times for varying input sizes. Note: The AND gates are predictions based on the average ratio of AND to total gates. The AND gates make up the total cost because of free XOR and NOT gates.

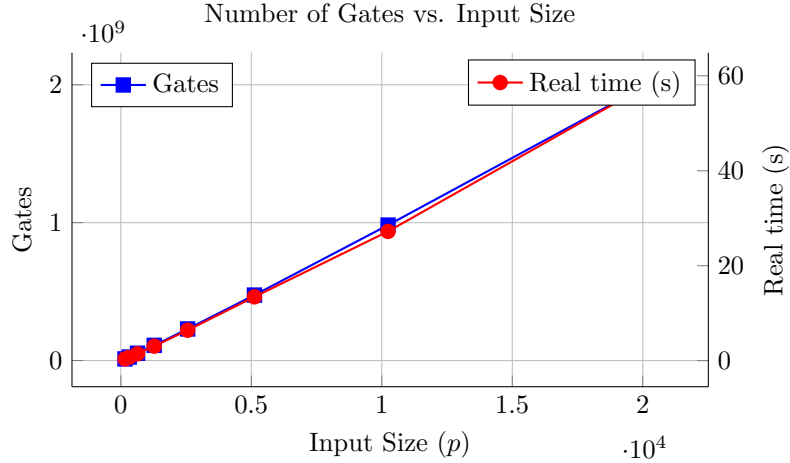


Figure 3: Number of gates and real execution time vs input size. As the plot shows, the number of gates and times in seconds correspond almost exactly to each other.

The cost in our implementation indeed grows according to $k \log N$, but the linear component is still significant. Further code optimization would improve

performance and would make the performance match the asymptotics more closely.

References

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